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A An Improved Model of the Human Feedback Process

As illustrated in Equation (1), the feedback process in RLHF is typically modeled with a single human $\mathcal{H}$ with internal reward function $r_{\mathcal{H}}$; examples sampled from the base model: $x_i \sim \pi_\theta$; and feedback as a function of the human, example, and noise: $y_i = f(h, x_i, \epsilon_i)$. However, as discussed in Section 3, this is a misspecified model of the process: there is not a single human, humans values are not representable with a reward function, human actions are dependent on context, and the sampling process can involve a human. Thus we propose an alternative formulation.

Let $\Delta\mathcal{H}$ refer to a joint distribution of humans (or groups thereof if feedback is provided collaboratively) used for obtaining samples and feedback denoted as $\mathcal{H}_j^{\text{sample}}$ and $\mathcal{H}_j^{\text{feedback}}$. A dataset of examples is sampled from π_θ (or some other source) where each example x_i is defined to be a batch of one or more generations from the base model. Importantly, x_i may not contain all information about the world state (e.g., if x_i is a 2D rendering of a 3D environment), and the human may be able to observe more than just the model's output (e.g., if interpretability tools are used to aid in evaluation). So let v be a rendering function that maps π_θ and x_i to what a human sees. The behavior of humans varies over time and in different contexts, so let c_i^{sample} and c_i^{feedback} represent particular contexts for sampling and feedback collection. Denote the sampling process as s which maps the base model π_θ, a human $\mathcal{H}_j^{\text{sample}}$, and context c_i^{sample} to some example x_i. Notably, s could ignore the base model and generate offline samples from some other source. Finally, let f map a human $\mathcal{H}_j^{\text{feedback}}$, rendered example $v(\pi_\theta, x_i)$, and context c_i^{feedback} to feedback y_i. The data collection process can thus be more completely modeled as:

$$\mathcal{H}_j^{\text{sample}}, \mathcal{H}_j^{\text{feedback}} \sim \Delta\mathcal{H}, \qquad x_i \sim s(\pi_\theta, \mathcal{H}_j^{\text{sample}}, c_i^{\text{sample}}), \qquad y_i = f(v(\pi_\theta, x_i), \mathcal{H}_j^{\text{feedback}}, c_i^{\text{feedback}}) \tag{4}$$

which highlights a need for future work to better account for the aspects of this process that are commonly not accounted for when training systems with RLHF.

B Rationale for Why Challenges Were Categorized as Tractable or Fundamental

In Section 3, we categorize problems as **tractable** or **fundamental**. The key distinction between the two is that fundamental challenges are substantial enough that overcoming them would require a method that is no longer a form of RLHF. Although many of the fundamental problems we identify can be alleviated by improving how RLHF is approached, they could be fully addressed with RLHF. As a result, they should be either avoided by not using RLHF or compensated for by other safety measures. This distinction is soft, and some categories of challenges are marginal. Here, we briefly explain each categorization.

B.1 Problems from Section 3.1:

Tractable: Selecting representative humans and getting them to provide quality feedback is difficult: This can be addressed by studying and improving the selection and training of evaluators.

Tractable: Some evaluators have harmful biases and opinions: This can be addressed by studying and improving the selection and training of evaluators.

Tractable: Individual human evaluators can poison data: This can be addressed with improved evaluator selection and quality assurance measures.

Tractable: Humans make simple mistakes due to limited time, attention, or care: This is marginal because human mistakes can never fully be overcome. However, they can be addressed with improved working conditions and quality assurance procedures.

Tractable: Partial observability limits human evaluators: Human evaluators can be provided with all information available in the policy's observations (although representing this in an easily-comprehensible way may be challenging).

Fundamental: Humans cannot evaluate performance on difficult tasks well: Human intelligence and cognitive capacity are limited. Humans cannot be expected to properly evaluate the performance of